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ENGINEERING



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Savitribi Institute of Medical And Technical Sciences
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Department Of Artificial Intelligence & Data Science

PCA-Based District-Wise Drinking Water Quality Risk Index and Disease Outbreak Correlation System for India

CSA0904 - PROGRAMMING IN JAVA

BATCH - 5

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PROBLEM STATEMENT



- Drinking water quality varies across districts in India due to natural and human-made factors.
- Poor water quality contributes to water-borne diseases such as cholera, typhoid, diarrhoea, and hepatitis.
- Existing monitoring systems often analyze water quality and disease data separately.
- Large numbers of water quality parameters make analysis complex and time-consuming.
- There is a need for an intelligent system that generates a district-wise risk index and correlates it with disease outbreaks.



OBJECTIVES



- Collect and preprocess district-wise drinking water quality and disease outbreak data from reliable sources.
- Apply Principal Component Analysis (PCA) to generate a Drinking Water Quality Risk Index.
- Analyze the correlation between water quality risk and disease outbreaks.
- Visualize district-wise risk levels using charts and an interactive choropleth map.
- To Develop a decision-support system that helps government agencies and health authorities identify high-risk districts and prioritize interventions.

LITERATURE REVIEW

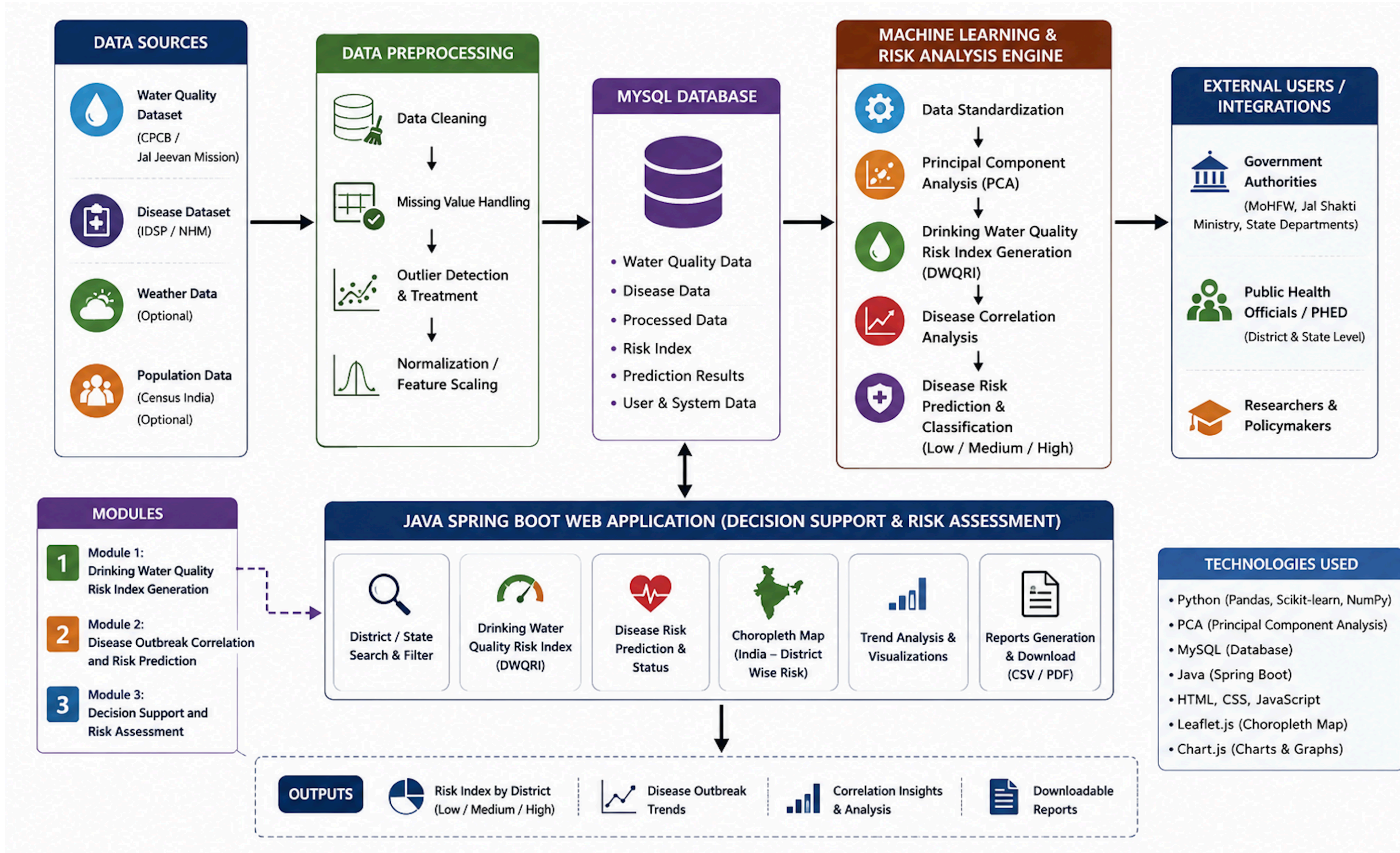
Year	Authors	Title	Methodology	Limitation
2022	K. R. N. Ashwin	Correlation, Regression Analysis, and Spatial Distribution Mapping of WQI for an Urban Lake in Noyyal River Basin in the Textile Capital of India	Computed Water Quality Index (WQI), performed correlation and regression analysis, and generated GIS-based spatial distribution maps to assess water quality.	Focused on a single urban lake; results cannot be generalized to district-level drinking water assessment across India.
2022	B. B. Ghosh et al.	Evaluation of Water Quality Index and Geochemical Characteristics of Surface Water from Tawang, India	Used PCA, Pearson correlation, ANOVA, and WQI to identify major pollution factors and evaluate surface water quality.	Study was limited to one region and focused on surface water rather than integrated drinking water sources.
2023	Biswanath Mahanty, Pema Lhamo, Naresh K. Sahoo	Inconsistency of PCA-Based Water Quality Index – Does It Reflect the Quality?	Evaluated the effectiveness of PCA-based WQI and analyzed its consistency in representing overall water quality.	Primarily examined methodological limitations of PCA-WQI and did not include disease correlation or spatial visualization.
2025	Gurudatta Singh, Supriya Chaudhary, Balendu Shekher Giri, Virendra Kumar Mishra	Assessment of Geochemistry and Irrigation Suitability of the River Ganga, Varanasi, India: PCA Reduction for Water Quality Index and Health Risk Evaluation	Applied PCA, WQI, hydrochemical analysis, and health risk assessment using multi-season water quality data.	Restricted to River Ganga (Varanasi); did not develop district-wise risk mapping or disease outbreak prediction.
2025	Yogeshwaran et al.	Assessment of Surface Water Quality Parameters Using Multivariate Analysis – A Case Study of Kurichi and Big Lakes in Coimbatore	Applied PCA, Factor Analysis, Cluster Analysis, Pearson correlation, and WQI to identify pollution sources and classify water quality.	Study focused on two lakes and lacked integration with public health or disease outbreak datasets.



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SYSTEM ARCHITECTURE





MODULE TITLES



Module 1: Drinking Water QRI Generation

Module 2: Disease Outbreak Correlation and Risk Prediction

Module 3: Decision Support and Risk Assessment



Module 1: Drinking Water QRI Generation



- Collect and preprocess district-wise drinking water quality parameters from reliable data sources.
- Normalize the dataset and remove inconsistencies to improve data quality and analysis accuracy.
- Apply Principal Component Analysis (PCA) to reduce dimensionality and identify the most influential water quality parameters.
- Generate a Drinking Water Quality Risk Index (DWQRI) for each district based on the PCA results.
- Classify districts into Low, Medium, and High risk categories for effective water quality assessment.



Module 2: Disease Outbreak Correlation and Risk Prediction



- Integrate district-wise disease outbreak data with the generated Drinking Water Quality Risk Index.
- Perform correlation analysis to examine the relationship between water quality and water-borne diseases.
- Analyze historical trends to identify patterns and factors contributing to disease outbreaks.
- Predict disease risk levels using the generated risk index and analytical models.
- Identify vulnerable districts and provide insights for early monitoring and preventive action.



Module 3: Decision Support and Risk Assessment



- Develop a user-friendly system for monitoring district-wise drinking water quality and disease risk.
- Display risk assessment results through interactive dashboards, charts, and choropleth maps.
- Enable state-wise and district-wise search, filtering, and comparison of risk levels.
- Generate reports summarizing water quality status, disease correlation, and risk classification.
- Support government agencies and public health authorities in making informed decisions for effective water quality management and disease prevention.



EVALUATION METRICS



- PCA explained variance ratio to measure information retained after dimensionality reduction.
- Correlation coefficient (Pearson/Spearman) to evaluate the relationship between water quality and disease outbreaks.
- Mean Squared Error (MSE) to assess prediction accuracy (if a prediction model is used).
- Root Mean Squared Error (RMSE) to measure prediction error.
- R^2 Score (Coefficient of Determination) to evaluate the performance of the prediction model.



EXPECTED OUTCOMES



- Generation of a district-wise Drinking Water Quality Risk Index.
- Identification of districts with low, medium, and high water quality risk.
- Discovery of correlations between poor water quality and disease outbreaks.
- Interactive dashboard with graphs and a choropleth map for easy visualization.
- A decision-support system that assists government agencies and public health authorities in monitoring water quality.



CONCLUSION



- The proposed system provides a comprehensive assessment of district-wise drinking water quality.
- PCA simplifies complex water quality data while preserving essential information.
- Correlation analysis helps identify links between water quality and disease outbreaks.
- Interactive visualizations improve understanding of geographical risk patterns.
- The system supports informed decision-making for better water resource management and public health planning.



THANK YOU!

ANY QUESTIONS?